

23CTR602 AGILE SOFTWARE DEVELOPMENT

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PRE-REQUISITES: Nil

OBJECTIVES:

- To acquire fundamental knowledge of Agile methodologies and practices
- To understand the various Agile frameworks like Scrum, XP, and Kanban and their application in real- world software development
- To apply Agile principles and tools to design, implement, and evaluate software systems

OUTCOMES:

Learners will be able to

CO1:	explain the fundamental principles and methodologies of Agile development	K2
CO2:	analyze different Agile frameworks such as Scrum and Extreme Programming (XP)	K4
CO3:	implement software projects using Agile methodologies like Scrum, XP, and Kanban	K3
CO4:	apply Agile project management techniques to monitor and adapt project progress	K3
CO5:	evaluate the effectiveness of Agile practices in improving software quality and team productivity	K5

UNIT I FUNDAMENTALS OF AGILE PROCESS

9

Introduction and background –Agile Manifesto and Principles – Stakeholders and Challenges – Overview of Agile Development Models: Scrum – Extreme Programming (XP) – Feature Driven Development –Crystal, Kanban – and Lean Software Development.

UNIT II AGILE PROJECTS

9

Planning for Agile Teams: Scrum Teams – XP Teams, General Agile Teams – Team Distribution; Agile Project Lifecycles: Typical Agile Project Lifecycles – Phase Activities, Product Vision – Release Planning: Creating the Product Backlog – User Stories, Prioritizing and Estimating – Creating the Release Plan; Monitoring and Adapting: Managing Risks and Issues – Retrospectives.

UNIT III SCRUM

9

Agile Scrum Framework – Scrum Artifacts – Meetings – Activities and Roles – Scrum Team Simulation – Scrum Planning Principles –Product and Release Planning –sprinting: Planning, Execution–Review and Retrospective; User story definition and Characteristics –Acceptance tests and Verifying stories – Burn down chart –Daily scrum –Scrum Case Study.

UNIT IV EXTREME PROGRAMMING (XP)

9

XP Lifecycle –The XP Team –XP Concepts: Refactoring –Technical Debt –Time boxing –Stories –Velocity; Adopting XP: Pre-requisites – Challenges; Applying XP: Thinking – Pair Programming –Collaborating – Release, Planning – Development; XP Case Study.

UNIT V AGILE SOFTWARE DESIGN AND DEVELOPMENT

9

Agile design practices – Role of design Principles – Need and significance of Refactoring – Refactoring Techniques, Continuous Integration –Automated build tools –Version control; Agility and Quality Assurance: Agile Interaction Design –Agile approach to Quality Assurance –Test Driven Development –Pair programming: Issues and Challenges.

TOTAL:45

TEXTBOOKS:

1. Johnny Schneider,” Understanding Design Thinking, Lean and Agile”, O’Reilly Media,2017.
2. Robert C. Martin, Agile Software Development- Principles, Patterns and Practices, Prentice Hall, 2013.
3. Kenneth S. Rubin, Essential Scrum: A Practical Guide to the Most Popular Agile Process, Addison Wesley, 2012

REFERENCE BOOKS:

1. James Shore and Shane Warden, “The Art of Agile Development”, O’Reilly Media, 2007.
2. Craig Larman, “Agile and Iterative Development: A manager’s Guide”, Addison-Wesley, 2004.
3. Ken Schawber, Mike Beedle, “Agile Software Development with Scrum”, Pearson, 2001.
4. Cohn, Mike, “Agile Estimating and Planning”, Pearson Education, 2006.
5. Cohn, Mike,” User Stories Applied: For Agile Software Development” AddisonWesley,2004.

WEB URLS:

1. <https://www.coursera.org/learn/agile-software-development>
2. <https://nptel.ac.in/courses/106105182>
3. <https://nptelvideos.com/video.php?id=904>

COURSE ARTICULATION MATRIX

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	-	-	1	-	-	-	1	1	-	1	1
CO2	2	3	-	-	1	-	-	-	1	1	-	1	1
CO3	3	3	2	2	1	-	-	-	1	1	-	1	1
CO4	3	3	2	2	1	1	1	-	1	1	-	1	1
CO5	3	3	2	2	1	1	1	-	1	1	-	1	1
CO	2.6	2.8	2	2	1	1	1	-	1	1	-	1	1